Exercise 3

1. SELECT
    product-name,
    price,
CASE
    WHEN price > 1000 THEN 'Expensive'
    WHEN price BETWEEN 100 AND 1000 THEN 'mid-range'
    ELSE 'Budget'
END AS 'price-Category'
FROM task.dbo.products;

Output

product-name	Price	price-Category
Laptop	1200.00	Expensive
Phone	800.00	Mid-range
Keyboard	45.00	Budget
Monitor	300.00	Mid-range
Mouse	25.00	Budget

2.

```
SELECT
    Customer-name,
    amount,
CASE
  WHEN amount >= 1000 THEN 'High value'
  WHEN amount BETWEEN 500 AND 999,99 THEN 'Medium value'
ELSE 'low value'
END AS 'order-value-category'
FROM task.dbo.orders;
```

Output

customer_name	Amount	Order-value-Category
Alice	150.00	low value
Bob	560.00	Medium value
Charlie	999.99	Medium value
Diana	45.50	low Value
Ethan	120.00	High value

3.

```
SELECT
    emp-name,
    department,
    Salary,
CASE
    WHEN department = 'IT' AND Salary > 80000 THEN 'Senior IT'
    WHEN department = 'HR' AND Salary > 55000 THEN 'Experienced HR'
ELSE 'Staff'
END AS 'position_level'
FROM task.dbo.employees;
```

Output

emp-name	Department	Salary	position Level
John	IT	85000.00	Senior IT
Sara	HR	60000.00	Experienced HR
mark	IT	75000.00	Staff
Lucy	Finance	95000.00	Staff
Tom	HR	55000.00	Staff

4.

```
SELECT
    Student_name,
    Score,
CASE
 WHEN score >= 90 THEN A
 WHEN score BETWEEN 80 AND 89 THEN 'B'
 WHEN score BETWEEN 70 AND 79 THEN 'C'
 WHEN score BETWEEN 60 AND 69 THEN 'D'
 ELSE 'F'
 END AS 'grade'
 FROM test.dbo.students;
```

Output

Student_name	Score	grade
Anna	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

5.

```
SELECT
    delivery_id,
    delivery_time_minutes,
CASE
  WHEN delivery_time_minutes <=30 THEN 'Fast'
  WHEN delivery_time_minutes BETWEEN 31 AND 60 THEN 'On Time'
ELSE 'late'
END AS 'performance'
FROM test.dbo.deliveries ;
```

<u>Output</u>

Delivery_id	Delivery_time_minutes	Performance
1	45	on Time
2	80	late
3	30	fast
4	65	late
5	100	late

6.

```
SELECT
    issue_type,
    priority,
CASE
  WHEN priority =3 THEN 'High'
  WHEN priority = 2 THEN 'Medium'
  WHEN priority = 1 THEN 'low'
END AS 'priority_label'
FROM test.dbo.tickets;
```

Output

~~Log~~ issue_type	priority	priority_label
login issue	1	low
Server down	3	High
Slow system	2	medium
Email error	2	medium
password reset	1	low

7.

```
SELECT
    Student_id,
  (days_present/total_days) *100 As attendance_percentage,
CASE
  WHEN (days_present/total_days) * 100 >= 90 THEN 'Excellent'
  WHEN (days_present/total_days) * 100 BETWEEN 75 AND 89 THEN 'Good'
  ELSE 'Needs improvement'
END As 'Attendance_status
FROM ~~task.dbo.attendance~~; task.dbo.attendance;
```

Output

Student_id	attendance_percentage	attendance_status
1	90	Excellent
2	60	Needs improvement
3	96	Excellent
4	50	Needs improvement
5	100	Excellent

8.

```
SELECT
  product_id,
  stock_qty,
CASE
  WHEN stock_qty >5 THEN 'IN stock'
  WHEN stock_qty BETWEEN 1 AND 5 THEN 'low stock'
ELSE 'out of stock'
END AS 'stock status'
FROM task.dbo.products_inventory;
```

Output

product_id	stock_qty	stock status
1	5	low stock
2	0	out of stock
3	25	in stock
4	10	in stock
5	3	low stock

9.

```
SELECT
  subject,
  enrolled_students,
CASE
  WHEN enrolled_students >= 25 THEN 'Large'
  WHEN enrolled_students BETWEEN 10 AND 24 THEN 'medium'
ELSE 'small'
END AS 'class_size_category'
FROM ~~dbo.classes~~; task.dbo.classes;
```

Output

Subject	Enrolled_students	Class_size_category
math	30	Large
English	25	large
science	15	medium
Art	5	small
History	20	medium

10.

```
SELECT
  payment_id,
  payment_method,
  amount,
CASE
WHEN amount >=200 AND payment_method = 'Cash'
 THEN 'Elligible For discount'
ELSE 'Not Elligible'
END AS 'Discount_eligibility'
FROM task.dbo.payments ;
```

Output

payment_id	payment_method	Amount	discount_eligibility
1	Card	50.00	Not Eligible
2	Cash	200.00	Elligible For discount
3	Card	150.00	Not Elligible
4	paypal	75.00	Not Eligible
5	cash	300.00	Eligible For discount